6G Weekly Survey — Oct 12 - Oct 18, 2026

15 items across 4 topics, ranked by importance.

6G Demos & Trials

gr-ntnisac: An Open GNU Radio Testbed for NTN-ISAC

Soham Dhiren Desai · arXiv eess.SP · 2026-10-08

★★★★■ Addresses a critical gap in NTN-ISAC experimentation by providing repeatable, controlled validation before costly satellite trials; strong engineering contribution with open-source release from a credible research context.

This paper presents gr-ntnisac, an open-source GNU Radio testbed for evaluating passive radar sensing using LEO satellite 5G NR downlink signals. The testbed emulates LEO channel conditions (tens of kHz Doppler, time-varying delays) with cabled USRPs and ground-truth path parameters for controlled validation. Initial experiments achieve 34.8 dB direct-path cancellation on hardware and demonstrate detection of a -25 dB moving target in 77% of observation intervals.

Contribution: First open testbed enabling reproducible, ground-truth validated development of ISAC receivers for NTN scenarios with realistic LEO Doppler and delay dynamics.

Relevance to your work: Directly enables reproducible research in two of your core areas (NTN and ISAC), providing a validated platform for testing AI/ML receivers under realistic LEO satellite channel conditions before deployment.

Noncoherent MIMO in the FR3 Upper Mid-Band Using Grassmannian Signaling: Prototype and Experimental Evaluation

Álvaro Pendás-Recondo, Enrique Pendás-Recondo, Álvaro F. Vaquero, Marcos R. Pino, et al. · arXiv eess.SP · 2026-10-06

★★★■■ Solid experimental validation of a known modulation technique in a new frequency regime relevant to 6G, though the noncoherent approach targets niche use cases (low spectral efficiency, high reliability) rather than mainstream scenarios.

The authors built and tested the first FR3 (15 GHz) hardware prototype of noncoherent MIMO using Grassmannian signaling, where neither transmitter nor receiver needs CSI. The system integrates unitary space-time modulation with OFDM on SDR platforms with custom RF front-ends, validated through over-the-air experiments in LoS/NLoS scenarios. Symbol error rate measurements match simulations, demonstrating practical feasibility for low-spectral-efficiency, high-reliability 6G applications.

Contribution: First experimental demonstration of Grassmannian signaling above 6 GHz in the FR3 upper mid-band, proving CSI-free MIMO operation is viable at 15 GHz with practical hardware.

Relevance to your work: Demonstrates an alternative to learned/AI-based CSI acquisition for FR3 MIMO; the prototype methodology and OTA validation approach could inform your work on neural receivers or channel-blind learning schemes.

RAPID: Proactive Rate Control for Enabling High Data Rate and Low Latency in 5G Edge Networks

Hamid Hassani, Sonia M. Heemstra de Groot, Ignas G. Niemegeers, Kishor C. Joshi, et al. · arXiv cs.NI · 2026-10-09

★★★■■ Solid applied contribution addressing a practical problem (latency vs throughput tradeoff) in private 5G networks for XR/tactile Internet use cases, though the approach is domain-specific transport-layer engineering rather than fundamental PHY or air-interface innovation.

This paper presents RAPID, a transport-layer rate control protocol for 5G edge networks that proactively regulates transmission rate to maintain target physical resource block (PRB) utilization at the base station. The approach aims to achieve simultaneously high data rate and low latency by avoiding buffer buildup. Testbed experiments with multi-UE scenarios show RAPID achieves 55-72x lower one-way delay versus

TCP Cubic with 16-27% goodput reduction and minimal packet loss.

Contribution: A proactive PRB-utilization-based rate control mechanism that operates at the transport layer to co-optimize latency and throughput in 5G edge deployments, validated on real hardware.

Relevance to your work: Tangential—this is upper-layer congestion control for application QoS rather than PHY-layer design, though it demonstrates real-world latency challenges in 5G edge that AI-native air interfaces and scheduling might address more fundamentally.

Blind and Asynchronous Initial Beam Acquisition and Clock-Assisted Re-Acquisition in Standalone Mobile MIMO-OFDM Systems

Ali Abdi, Firooz B. Saghezchi, Haris Gacanin · arXiv eess.SP · 2026-10-09

★★★■■ Solid incremental work addressing beam management with a tested prototype, but the approach is conventional sweeping-based rather than AI/ML-driven or fundamentally novel for 6G.

The paper presents a beam-management framework for mobile MIMO-OFDM combining Blind Initial Beam Acquisition (BIBA), which requires no prior synchronization or feedback channel, with Clock-Assisted Re-Acquisition (CARA) that exploits established clock sync to reduce re-acquisition time by ~55%. The protocols are validated on a 28 GHz mmWave testbed with realistic hardware constraints (beam-switching latency, TDD timing, SDR processing delays) and formalized via a unified finite state machine model. Probabilistic feasibility bounds are derived for random UE mobility scenarios.

Contribution: A practical, asynchronous beam acquisition protocol validated on real mmWave hardware that eliminates the need for prior synchronization and out-of-band feedback, with a clock-assisted fast re-acquisition mechanism.

Relevance to your work: Directly relevant to beam management challenges in massive MIMO and mmWave systems, though the classical sweeping approach contrasts with AI-native methods (learned codebooks, neural beam prediction) central to your beyond-5G research.

AI Models Can Provably Hide Arbitrary Capabilities

Andis Draguns, Sumeet Ramesh Motwani, Raymond Douglas, Christian Schroeder de Witt · arXiv cs.LG · 2026-10-09

★★■■■ While technically rigorous on AI safety/security, this adversarial ML work has no direct bearing on physical layer design, air interface protocols, or wireless system architecture for 6G.

The authors demonstrate that AI models can cryptographically hide arbitrary capabilities using backdoor attacks with encrypted neural networks embedded in model weights, activated only by secret triggers. They implement proof-of-concept models that resist standard elicitation methods (fine-tuning, gradient search, adversarial training) on a chemical reaction prediction task. The work establishes fundamental limits on capability evaluation and model auditing under adversarial conditions with cryptographic guarantees.

Contribution: First provably hard-to-elicit backdoor construction that hides arbitrary computations (not just fixed outputs) in neural network weights under cryptographic assumptions, extending beyond previous string-hiding attacks.

Relevance to your work: tangential

SeoulMMOD: A Large-Scale Multimodal Origin-Destination Flow Benchmark

Taeyoung Yu, Seonbin Jo, Jiwon Kim, Junyoung Byun · arXiv cs.LG · 2026-10-09

★★■■■ While this is competent urban-mobility work with a useful dataset, it addresses transportation forecasting rather than wireless communications and has no direct connection to air interface design, channel modeling, or spectrum management for 6G.

The paper introduces SeoulMMOD, a three-year hourly origin-destination flow dataset covering six transport modes across Seoul's 25 districts and 426 sub-districts, with auxiliary signals (travel time, POI, weather). Benchmarking 14 forecasting models reveals that district-level spatiotemporal methods fail to scale to finer sub-district resolution and that joint multi-mode training does not consistently outperform mode-separate training. The dataset provides a reproducible testbed for citywide multimodal mobility forecasting with

cross-year generalization experiments.

Contribution: First large-scale public benchmark combining multiple transport modes at fine spatial granularity over multiple years, exposing scaling limitations of existing spatiotemporal forecasting architectures and lack of generalization benefits from naive joint multi-mode training.

Relevance to your work: tangential

Secrets in Preamble: Practical and Robust PHY Information Hiding for Wi-Fi

Rui Pan, Guanxiong Shen, Linning Peng, Xinyu Qi, et al. · arXiv eess.SP · 2026-10-09

★★■■■ Solid demonstration of PHY-layer information hiding with hardware validation, but the application domain (steganography/watermarking) is orthogonal to core 6G research priorities like spectral efficiency, AI-native air interfaces, or sensing integration.

The authors propose embedding covert information in Wi-Fi legacy short training field (L-STF) preambles using neural networks that learn optimal embedding rules while preserving autocorrelation properties required for packet detection and frequency offset compensation. They deploy a hardware prototype demonstrating 46 secret bits per packet with robust extraction across indoor and outdoor channels. Experimental validation confirms practical feasibility for PHY-layer steganography and watermarking applications.

Contribution: First practical neural-network-based approach to embed secrets in Wi-Fi L-STF that maintains standard receiver functionality while achieving significantly higher capacity (46 bits/packet) and environmental robustness than prior Wi-Fi information hiding methods.

Relevance to your work: Tangential—while this uses neural networks for PHY waveform manipulation and demonstrates learned encoding strategies, the covert communication use case does not align with your focus on AI-native receivers, channel modeling, CSI compression, or ISAC for next-generation wireless systems.

HPQ-AKE: A Provably Secure Sign-Less Hybrid Authenticated Key Exchange Protocol for Bandwidth-Constrained IoT and Edge Networks

Khiem Pham-Tuan, Minh Quang Le, Khuong Nguyen-An · arXiv cs.NI · 2026-10-08

★★■■■ Competent cryptographic engineering for IoT security, but narrowly focused on post-quantum migration rather than 6G physical-layer or air-interface innovation.

The paper presents HPQ-AKE, a hybrid authenticated key exchange protocol for IoT/edge networks that combines ML-KEM-768 (post-quantum) with RSA-OAEP to avoid post-quantum signatures, reducing handshake data by 56.4% (13,009→5,668 bytes) compared to hybrid TLS 1.3. The protocol is proven secure under extended Bellare-Rogaway models and evaluated via x86 benchmarking plus Monte Carlo simulation showing 31.4% latency reduction on satellite-like links (50 kbps, 600 ms RTT). Break-even bandwidth thresholds are 0.31 Mbps (pre-cached) and 4.08 Mbps (full handshake) at 20 ms RTT.

Contribution: A sign-less dual-KEM construction that eliminates post-quantum signature overhead during authentication, formally proven secure while maintaining backward compatibility with RSA infrastructure.

Relevance to your work: Tangential—addresses link-layer/application-layer security for bandwidth-constrained satellite/NTN scenarios but does not engage with channel modeling, AI-native PHY, or wireless-specific resource allocation.

Moving Target Defense in SDN-enabled EV Charging Network

Roland Plaka, Mikael Asplund, Simin Nadjm-Tehran · arXiv cs.NI · 2026-10-08

★■■■■ While competently executed for critical infrastructure security, this work is entirely focused on network-layer cybersecurity for EV charging systems and has no connection to wireless PHY, air interface design, or radio resource management relevant to 6G research.

This paper addresses low-rate DoS attacks on SDN-enabled EV charging infrastructure by proposing CS-SHIELD, a moving target defense that detects malicious flow-table rules via cross-layer identity verification and responds by dynamically reassigning virtual IPs to active chargers. Experiments on an

emulated SDN testbed with real charging protocols show the mechanism maintains full site availability under attack with negligible overhead. The IP shuffling invalidates attacker reconnaissance, preventing exhaustion of SDN switch flow tables.

Contribution: A practical moving target defense specifically tailored to SDN-enabled EV charging networks that combines cross-layer verification with dynamic IP reassignment to counter low-rate DoS attacks targeting flow tables.

Relevance to your work: tangential

6G Physical Layer

Statistical Analysis of 3D Ambiguity Functions for Random MIMO-OFDM Signals

Pinjun Zeng, Yuyang Lu, Jialan Tang, Xiaoyang Chen, et al. · arXiv eess.SP · 2026-10-08

★★★★■ Strong analytical contribution addressing a fundamental ISAC challenge (random payload data affecting radar performance) with novel 3D spatial treatment relevant to massive MIMO and beamforming-based 6G sensing.
This paper derives a closed-form statistical characterization of the 3D ambiguity function (delay, Doppler, angle) for random MIMO-OFDM waveforms used in ISAC systems. The analysis expresses the ambiguity function as a quadratic form of random communication symbols and shows how constellation kurtosis affects sidelobe levels. The framework extends to multi-symbol coherent processing with fast-slow time analysis for improved Doppler resolution.

Contribution: First rigorous statistical analysis of MIMO-OFDM ambiguity functions incorporating spatial (angular) domain alongside delay-Doppler, with explicit characterization of how constellation properties and MIMO precoding affect sensing performance.

Relevance to your work: Directly relevant to ISAC research interests, particularly for understanding fundamental sensing-communication tradeoffs in AI-native air interfaces where adaptive modulation and MIMO precoding must coexist with radar functionality.

2D Coprime Pilots for Delay-Doppler Sensing in OFDM-ISAC Systems

Guanping Shang, Zeyan Zhuang, Anzheng Tang, Shenghui Song, et al. · arXiv eess.SP · 2026-10-08

★★★★■ Strong theoretical contribution addressing a fundamental ISAC design problem with practical implications for 6G pilot overhead, from credible research group using rigorous mathematical framework.
The paper addresses joint optimization of sensing range and accuracy in OFDM-ISAC systems using sparse pilot patterns. The authors derive a 2D coprime condition (time-frequency) as the first necessary and sufficient criterion for unambiguous delay-Doppler estimation, then propose an Adam-Guided Iterative Refinement algorithm to maximize sensing mutual information while minimizing pilot overhead. Numerical results show improved performance over existing pilot designs by exploiting the flexibility of the 2D coprime framework.

Contribution: First necessary and sufficient condition for unambiguous delay-Doppler estimation in OFDM-ISAC using 2D coprime pilots, unified with an SMI-based optimization framework for joint range-accuracy maximization.

Relevance to your work: Directly relevant: provides principled design for OFDM-ISAC pilot patterns balancing communication overhead and sensing performance, applicable to AI-native air interface design and integrated sensing-communication architectures.

Hierarchical RIS Association for FR3 Space-Air-Ground Integrated Networks

Muddasir Rahim, Irfan Azam, Soumaya Cherkaoui · arXiv eess.SP · 2026-10-06

★★★■■ Solid incremental work combining RIS with NTN/SAGIN in FR3, but association optimization via matching is a standard technique and the system model lacks implementation details (channel models, RIS phase optimization, FR3 propagation).

Proposes a hierarchical RIS association scheme for FR3 space-air-ground integrated networks where terrestrial base stations serve primary users via terrestrial RISs, while LEO satellites provide backup connectivity via UAV-mounted aerial RISs. Formulates sum-rate maximization as a mixed-integer nonconvex problem with binary association variables and RIS capacity constraints, solved via a many-to-one matching framework that prioritizes terrestrial links before satellite fallback. Simulations show significant sum-rate gains over greedy and random association baselines.

Contribution: First hierarchical matching framework for jointly optimizing terrestrial and aerial RIS associations in FR3 SAGIN systems, explicitly prioritizing terrestrial connectivity before invoking satellite backup links.

Relevance to your work: Directly relevant to your NTN research agenda—combines LEO satellites with terrestrial networks and addresses FR3 spectrum for 6G, though the RIS association problem is more operational than physical-layer focused.

RIS-Assisted Reliability Maximization for URLLC in Upper Mid-Band 6G Networks

Muddasir Rahim, Irfan Azam, Soumaya Cherkaoui · arXiv eess.SP · 2026-10-06

★★★■■ Solid incremental work combining known techniques (SCA, matching theory) for a relevant 6G scenario (FR3 + RIS + URLLC), but lacks fundamental novelty in methods or channel modeling.

This paper addresses reliability maximization for URLLC in 6G FR3 (upper mid-band, 7–24 GHz) networks using reconfigurable intelligent surfaces (RIS). The authors formulate a joint power allocation and UE-RIS association problem under finite blocklength constraints and solve it via successive convex approximation combined with matching theory. Simulations demonstrate near-optimal performance compared to exhaustive search and significant gains over greedy/random baselines.

Contribution: First joint optimization of power and RIS association specifically for URLLC reliability in FR3 bands under finite blocklength regime, using a computationally tractable multiphase framework.

Relevance to your work: RIS-assisted networks are relevant to your beyond-5G research, and the finite blocklength formulation connects to physical-layer reliability optimization, though the optimization approach itself does not leverage AI/ML methods you focus on.

Channel modeling / GAN-based

From Echoes to Evidence: Cooperative Fusion for Integrated Sensing and Communication

Xingqin Lin · arXiv cs.NI · 2026-10-07

★★★★■ Strong practical contribution with rigorous channel modeling (3GPP + ray tracing), substantial performance gains in realistic scenarios, and credible methodology addressing key ISAC deployment challenges like false targets and weak-echo exploitation.

This paper addresses cooperative ISAC for UAVs and mobile platforms, where multipath creates virtual targets and weak echoes are underutilized. The authors develop a fusion framework separating association confirmation, state refinement, and residual recovery, using a 3GPP channel model and San Francisco ray-tracing validation. Controlled experiments show fusion increases detection from 88.75% to 92.46% and reduces false outputs from 11.06% to 0.40%, with 90th-percentile position error improving from 8.54m to 3.50m.

Contribution: A minimax estimation framework for cooperative ISAC fusion that handles multipath ambiguities, shared-reflector confusion, and correlated errors across monostatic/bistatic links with explicit association controls.

Relevance to your work: Directly relevant: demonstrates end-to-end cooperative ISAC system performance under realistic 3GPP channel models, bridging sensing-communication fusion with channel modeling and providing UAV/vehicular benchmarks for JCAS architectures.

ISAC (sensing + communication)

Multistatic Terahertz ISAC for Object Shape Identification

Ralph Kumah Assan, Suresh Singh · arXiv eess.SP · 2026-10-08

★★★★■ This provides rare sub-THz ISAC system-level modeling with physics-grounded forward models and practical insights on transmitter diversity requirements, directly relevant to 6G spectrum discussions and commercially viable ceiling-mounted infrastructure—though limited to shape classification rather than full 3D reconstruction.

The paper investigates passive object shape identification in 6G ISAC by analyzing terahertz (300 GHz, 50 GHz BW) echoes from user equipment transmissions scattering to ceiling-mounted access points, using no dedicated sensing hardware. A physics-based multistatic forward model combining Physical Optics, Physical Theory of Diffraction, multipath modeling, and molecular absorption reveals that spectral slope alone cannot determine object geometry because six physically distinct scattering mechanisms share only three frequency exponents, making angular diversity mandatory for identifiability. Comparing a transmitter-count-invariant 2D-CNN against a physics-informed RBF-SVM on unseen orientations, accuracy rises from ~70% with one transmitter to 96-97% with 20, with the CNN excelling under irregular placement and lossy targets while the SVM matches performance on metal objects under quasi-uniform placement.

Contribution: The work rigorously demonstrates that terahertz ISAC shape classification fundamentally requires angular diversity because scattering mechanism degeneracy prevents geometry inference from spectral features alone, and validates multistatic sensing performance with minimal (4+) transmitter deployments achieving near-saturation detection in multi-object interference-limited scenarios.

Relevance to your work: Directly addresses passive ISAC at sub-THz frequencies using communication waveforms, providing angular diversity requirements and performance bounds highly relevant to joint sensing-communication PHY design and next-generation ISAC waveform development in your research portfolio.